HG2J LSD Registers and Functions

HMI special data registers supported for HG2J in the WindO/I-NV4 manual. Addresses are decimal. Feature-specific entries require the relevant hardware and project settings. Reserved ranges are listed at the end.

Timing and current clock

Register	Function and values
LSD4	Maximum scan time in milliseconds; reset when the base screen switches.
LSD5	Screen switching response time in milliseconds.
LSD6	External Device Communication 1 read scan time in milliseconds; stores the maximum and resets when the base screen switches.
LSD7	Scan counter; increments once per scan.
LSD8	One-second counter.
LSD9	10 ms counter.
LSD10	100 ms counter.
LSD11	200 ms counter.
LSD12	500 ms counter.
LSD13	Current clock year; four BCD digits.
LSD14	Current clock month; two BCD digits.
LSD15	Current clock day; two BCD digits.
LSD16	Current clock hour; two BCD digits.
LSD17	Current clock minute; two BCD digits.
LSD18	Current clock second; two BCD digits.
LSD19	Current day of week, one BCD digit: 0 Sunday through 6 Saturday.
LSD27	Current scan time in milliseconds.
LSD28	Minimum scan time in milliseconds.
LSD29	SNTP result: 0 success, 2 timeout, 4 other error.
LSD30	Minutes since the last successful SNTP time update, 0–65535. Resets on success; a failed request does not reset it.

Counters LSD7–12 wrap from 0xFFFF to 0. Timing values LSD4–6 have a stated error of ±10 ms. The manual gives the HG2J clock limit as 2038-01-19 03:14:07.

Clock setting and screen control

Register	Function and values
LSD20	Write 1 to apply the clock settings in LSD21–26. Returns to 0 after updating. Invalid date or time data prevents the update.
LSD21	Clock setting year; two BCD digits.
LSD22	Clock setting month; two BCD digits.
LSD23	Clock setting day; two BCD digits.
LSD24	Clock setting hour; two BCD digits.
LSD25	Clock setting minute; two BCD digits.
LSD26	Clock setting second; two BCD digits.
LSD31	Currently displayed screen number. BIN or BCD according to the project Screen No. Format setting.
LSD32	Write a base screen number to switch screens. BIN or BCD according to the project Screen Number Format setting.
LSD34–35	USB2 flash drive free capacity in KiB: LSD34 lower word, LSD35 upper word.
LSD36–37	USB2 flash drive total capacity in KiB: LSD36 lower word, LSD37 upper word.
LSD42	USB1 flash drive error: 0 normal; 1 write attempted with no drive or an incompatible drive; 2 format error; 3 access error, insufficient capacity, or read/write failure; 4 picture-data read failure.
LSD43–44	USB1 flash drive free capacity in KiB: LSD43 lower word, LSD44 upper word.
LSD45–46	USB1 flash drive total capacity in KiB: LSD45 lower word, LSD46 upper word.

Capacity values use 1024 bytes per KiB and round fractional KiB upward. For each pair, capacity = lower word + 65536 × upper word.

Alarms scripts display and user communication

Register	Function and values
LSD49	O/I Link slave station number; read only.
LSD50	Alarm List Display cursor selection: message sequence number, or alarm channel number. Range 1–1024; number = (block number − 1) × 16 + bit number + 1.
LSD51	Display brightness level. The manual specifies −16 to 31 for HG2J.
LSD52	ID of the script that caused an error.
LSD53	Script error: 0 normal; 1 processing; 2 execution time exceeded; 3 write count; 4 indirect device; 5 parameter; 6 cyclic script execution time exceeded; 7 fixed interval execution.
LSD56	Number of data lines from the start line to the cursor-selected line in Alarm List Display or Alarm Log Display.
LSD57	Number of alarm log records in the data storage area, 0–1024.
LSD65	Maximum screenshots saved to the USB1 flash drive: 1–999; default 99.
LSD67	Ethernet user communication TCP connection status: bit 0 User Communication 1; bit 1 User Communication 2; bit 2 User Communication 3. 0 disconnected, 1 connected.
LSD68	Force Ethernet user communication TCP disconnection on a 0-to-1 transition. Bits 0, 1 and 2 select User Communication 1, 2 and 3 respectively.
LSD72	Sound file number currently playing.
LSD73	Sound ID that could not be played because of an error.
LSD97	External Device Communication 2 read scan time in milliseconds.
LSD98	External Device Communication 3 read scan time in milliseconds.
LSD99	External Device Communication 4 read scan time in milliseconds.
LSD101	O/I Link slave polling period.
LSD102	O/I Link master slave registration settings.
LSD104	O/I Link master slave online information.
LSD106	O/I Link master slave error information.
LSD112	Communication driver register; meaning depends on the selected driver. See the External Device Setup Manual.

Ethernet email web and FTP

Register	Function and values
LSD120–123	Preferred DNS server IPv4 address, one octet per register. Apply with LSM53 rising from 0 to 1.
LSD124–127	Alternate DNS server IPv4 address, one octet per register. Apply with LSM53 rising from 0 to 1.
LSD192–195	HMI IPv4 address, one octet per register. Apply with LSM53 rising from 0 to 1.
LSD196–199	HMI subnet mask, one octet per register. Apply with LSM53 rising from 0 to 1.
LSD200–203	HMI default gateway IPv4 address, one octet per register. Apply with LSM53 rising from 0 to 1.
LSD204	External Device ID whose IP address and port will be read or changed.
LSD205–208	IPv4 address of the external device selected by LSD204, one octet per register. Apply with LSM54 rising from 0 to 1.
LSD209	Port number of the external device selected by LSD204. Apply with LSM54 rising from 0 to 1.
LSD213	HMI maintenance communication port number. Apply with LSM53 rising from 0 to 1.
LSD221	Number of email messages waiting to send.
LSD222	Email send result: 0 success; 1 parameter error; 2 timeout (30 seconds); 3 authentication error; 4 other error.
LSD223	Number identifying the email that has been sent.
LSD227	TCP service usage bitfield: bit 0 NV4 monitoring; bit 1 maintenance communication on its configured port; bit 2 pass-through (2101 or 2538). 0 unused, 1 in use. Bit 3 is FT2J/1J only.
LSD229	Custom web page string encoding: 0 UTF-8 without conversion; 1 Shift-JIS; 2 ISO 8859-1; 3 GB2312; 4 BIG5; 5 KSC5601; 6 ANSI1250; 7 ANSI1257; 8 ANSI1251.
LSD230	FTP server number selected by the HMI FTP client function.
LSD231	Successful FTP file transfer count. Reset at transfer start; increments after each successful file.
LSD232	Failed FTP file transfer count. Reset at transfer start; increments when a file transfer fails.

HMI Ethernet presets load from the project at power-up. Changing LSD204 loads that external device’s project IP address and port into LSD205–209.

Versions BACnet and wireless LAN

Register	Function and values
LSD240–241	System software version. Example: version 4.38 gives LSD240 = 0x0438 and LSD241 = 0x0000.
LSD242–243	WindO/I-NV4 version. Example: version 1.2.3 gives LSD242 = 0x0102 and LSD243 = 0x0300.
LSD260	BACnet/IP status bits: bit 0 stopped; bit 1 ready; bit 2 operating; bit 3 stopped by error.
LSD261	Last BACnet/IP error bits: bit 0 normal; bit 1 invalid device ID; bit 2 invalid IP address; bit 3 invalid BBMD IP address; bit 4 BBMD registration failure.
LSD286–291	MAC address, one byte per register in address order. Example: 00:03:7B:F0:12:A8 gives 0x0000, 0x0003, 0x007B, 0x00F0, 0x0012, 0x00A8.
LSD292–308	Wireless LAN SSID preset. Two characters per word; the 17th word is treated as a null terminator. Add a null terminator for a string shorter than 16 words. Apply with LSM77 rising from 0 to 1.
LSD309–341	Wireless LAN password preset. Two characters per word; the 33rd word is treated as a null terminator. Add a null terminator for a string shorter than 32 words. Apply with LSM77 rising from 0 to 1.
LSD344–347	Wireless LAN IPv4 address preset, one octet per register. Apply with LSM77 rising from 0 to 1.
LSD348–351	Wireless LAN subnet mask preset, one octet per register. Apply with LSM77 rising from 0 to 1.
LSD352–355	Wireless LAN default gateway preset, one octet per register. Apply with LSM77 rising from 0 to 1.
LSD356	Wireless LAN maintenance communication port preset. Apply with LSM77 rising from 0 to 1.
LSD366	Wireless LAN connection status: 0 no setting; 1 stopped; 4 connecting. These are the values listed in this manual.
LSD367	Signal strength for the SSID in LSD292–308: 0 no signal; 1 bad; 2 normal; 3 good. Cannot be measured for a hidden SSID.

Wireless LAN presets load from the project at power-up. Wireless LAN entries require the corresponding wireless LAN setup.

Other services and reserved addresses

Register	Function and values
LSD368	Number of social media messages waiting to send.
LSD369	Social media send result: 0 success; 1 parameter error; 2 timeout; 3 authentication error; 4 user error; 5 other error.
LSD370	Number identifying the social media message that has been sent.
LSD378	Accumulated hot-operation minutes inside the HMI. Increments every minute while the unit is hot; retained across power-off.
LSD390	Execution information for writing MQTT basic settings from an external memory device. Detailed values are in the Network User's Manual, MQTT Communication chapter.
LSD391	Execution error information for writing MQTT basic settings from an external memory device. Detailed values are in the Network User's Manual, MQTT Communication chapter.
LSD410	EtherNet/IP status: 0x0000 stopped; 0x0100 ready; 0x0200 executing; 0x0300 restarting; 0x0F00 stopped by error.
LSD411	Last EtherNet/IP error: 0x0000 normal; 0x0800 timeout waiting for link up; 0x8000 timeout waiting for communication to start; 0x8100 timeout waiting for communication to stop.

Reserved ranges

LSD0–3, 41, 47–48, 54–55, 60–64, 66, 69–71, 74–78, 80–96, 100, 103, 105, 107–111, 113–119, 128–154, 156–164, 166–191, 210–212, 214–220, 224–226, 228, 233–239, 244–247, 250–255, 257–259, 262–269, 278–285, 342–343, 357–365, 371–377, 379–389, 392–409 and 412–511.

Addresses not supported on HG2J

The manual marks LSD33, 38–40, 58–59, 79, 155, 165, 248–249, 256 and 270–277 as unsupported on HG2J. Their functions belong to other HMI models and are omitted here.

Source

IDEC WindO/I-NV4 Internal Device reference, R513a1403, dated April 27 2026. Source file IDEC_WINDOI_NV4_InternalDocs-1.pdf, PDF pages 1683–1703. Functions and HG2J support are taken from its HMI Special Data Registers table and notes. This document does not assign functions to undocumented or reserved addresses.